

Satyaghosh Maurya

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🔍 Google Scholar

Professional Summary

I am currently a postdoctoral researcher specializing in single-molecule techniques, biological membrane dynamics, and protein assembly. I am skilled in advanced microscopy methods, protein engineering, and single-molecule data analysis in MATLAB and Python. I am passionate about unraveling complex biological processes at the molecular level.

My current project aims to develop and apply advanced single-molecule fluorescence spectroscopy techniques coupled with nano plasmonics to directly observe and characterize transition paths in protein folding landscapes, potentially providing unprecedented insights into protein folding mechanisms and conformational dynamics.

Academic Details

Postdoctoral fellow

Department of Chemical and Biological Physics
Weizmann Institute of Science, Gilad Haran Lab

Rehovot, Israel

December, 2022 - present

Research Associate

Department of Chemical Engineering,
Indian Institute of Science (IISc)

Bangalore, India

April, 2022 - December, 2022

Ph.D.

Department of Chemical Engineering,
Indian Institute of Science (IISc)

Bangalore, India

August, 2015-March, 2022

Master's in Engineering (M.E.)

Department of Chemical Engineering,
Indian Institute of Science (IISc)

Bangalore, India

August, 2013-June 2015

Bachelor of Technology (B.Tech.)

Department of Chemical Engineering,
National Institute of Technology (NIT)

Durgapur, India

July, 2009-April 2013

Graduate Academic Projects

Assembly dynamics of pore-forming toxin ClyA using single-molecule techniques

- Understanding membrane protein dynamics and assembly of pore-forming protein Cytolysin A on supported lipid bilayers.

Biological membrane crowding by polymers

- Mobility and molecular self-assembly of biomolecules on polymer grafted lipid membranes.

Anti-microbial peptides

- Role of bioactive compounds in bacterial killing by LL37 anti-microbial peptide.

Teaching and Mentoring Experiences

Teaching assistant

Course: Chemical Engineering Mathematics

Indian Institute of Science, Bangalore

August, 2017 - December, 2017

Teaching assistant
Bangalore microscopy course

National Centre for Biological Sciences, Bangalore
September, 2018

Mentoring

- Junior graduate students (2)
- Project assistants (2)
- Summer interns (7)

Publications

- A Kulshrestha, **S Maurya**, T Gupta, R Roy, S Punnathanam, K.G Ayappa *"Conformational flexibility is a key determinant of the lytic activity of the pore forming protein, Cytolysin A"*, The Journal of Physical Chemistry B, 2023
- **S Maurya**, V H Rai, A Upasani, S Umrao, D Parwana, R Roy *Single molecule diffusion and assembly on polymer-crowded lipid membranes*, Journal of Visualized Experiments, 2022 ;
- S Jagtap, K Ratnasri, P Valloly, R Sharma, **S Maurya** et.al. *"Evaluation of spike protein antigens for SARS-CoV-2 serology"*, Journal of Virological Methods, 2021
- S Banerjee, **S Maurya**, Rahul Roy *"Single-Molecule Fluorescence Imaging: Generating Insights into Molecular Interactions in Virology"*, Journal of Biosciences, 2018
- P Sathyanarayana, **S Maurya**, A Behera, M Ravichandran, S S. Visweswariah, K.G Ayappa, R Roy *"Cholesterol Promotes Cytolysin A Activity by Stabilizing the Intermediates during Pore Formation"*, Proceedings of the National Academy of Sciences, (PNAS) 2018

Academic Conferences and Schools

- Single-Molecule Sensors and Nano Systems International Conference (S3IC), 2020, Online
- Seeing is Believing, EMBL/EMBO, poster titled "Investigating oligomerization pathway of pore forming toxin ClyA", 2019
- **Zeeshan Khan memorial award for best paper presentation** in Bangalore Microscopy Course, National Centre for Biological Sciences (NCBS), titled: "Cholesterol promotes Cytolysin A activity by stabilizing the intermediates during pore formation", 2018
- **Indian Biophysical Society Meeting**, Indian Institutes of Science Education and Research (IISER), Mohali, **best poster titled**, "Cholesterol promotes Cytolysin A activity by stabilizing the intermediates during pore formation", 2017
- National Workshop on Fluorescence and Raman Spectroscopy, Indian Institutes Technology (IIT), Guwahati, 2017
- Optics Within Life Science, Tata Institute of fundamental research (TIFR) and Indian Institute of Technology (IIT), Bombay, 2016
- Fluorescence and Raman Spectroscopy, Indian Institutes of Science Education and Research (IISER), Pune, 2015
- Winter School on Quantitative Systems Biology, International Centre for Theoretical Sciences (ICTS), Bangalore, 2015

Skills and Expertise

- **Advanced Microscopy:** Single-particle tracking (SPT), Single-molecule FRET (smFRET), FRAP, FCS, PALM, AFM, TEM
- **Molecular Biology:** Site-directed mutagenesis, Protein purification, Supported lipid bilayer preparation
- **Data Analysis:** MATLAB, Python, Image processing
- **Other Techniques:** Soft lithography, Microfluidics, Focused ion beam/Electron beam lithography